

$\xi > 0$ is the solution of $\kappa^{-1} = p^{-1} \sum_{i=1}^p (1 + (\xi \lambda_i)^{-1})^{-1}$,

$$\gamma = \frac{\sigma^2 + \sum_{i=1}^p \lambda_i \zeta_i^2 \bar{\beta}_i^2}{1 - \frac{\kappa}{p} \sum_{i=1}^p (1 + (\xi \lambda_i)^{-1})^{-2}},$$

$$\zeta_i = (1 + \xi \lambda_i)^{-1}, \quad \phi_i = \kappa \gamma (1 + (\xi \lambda_i)^{-1})^{-2}, \quad 1 \leq i \leq p.$$

The non-asymptotic distributional prediction is given by the following U -rotated normal distribution

$$\mathcal{D}_{\sigma, \Sigma, \beta} = \mathcal{UN}((\mathbf{1}_p - \zeta) \odot \bar{\beta}, p^{-1} \text{diag}(\lambda^{-1} \odot \phi)).$$

We remark that this definition is similar in spirit to the concurrent/recent work (?). However, unlike this work, here we prove the asymptotic correctness of the DC, we use it to rigorously predict the pruning performance and also extend this to retraining DC as discussed next.

Retraining DC. As the next step, we would like to characterize the DC of the solution after retraining, i.e., $\hat{\beta}^{RT}$. We carry out the retraining derivation (for magnitude pruning) as follows. Let $\mathcal{I} \subset [p]$ be the nonzero support of the pruned vector $\hat{\beta}_s^M$. Re-solving (3) restricted to the features over $\mathcal{I}$ corresponds to a linear problem with effective feature covariance $\Sigma_{\mathcal{I}}$ with support of non-zeros restricted to $\mathcal{I} \times \mathcal{I}$. For this feature covariance, we can also calculate the effective noise level and global minima of the population risk $\beta_{\mathcal{I}}^*$. The latter has the closed-form solution $\beta_{\mathcal{I}}^* = \Sigma_{\mathcal{I}}^\dagger \Sigma \beta^*$. The effective noise is given by accounting for the risk change due to the missing features via $\sigma_{\mathcal{I}} = (\sigma^2 + \beta^{*T} \Sigma \beta^* - \beta_{\mathcal{I}}^{*T} \Sigma_{\mathcal{I}} \beta_{\mathcal{I}}^*)^{1/2}$. With these terms in place, fixing $\mathcal{I}$ and using Def. 4, the retraining prediction becomes $\mathcal{D}_{\sigma_{\mathcal{I}}, \Sigma_{\mathcal{I}}, \beta_{\mathcal{I}}^*}$. This process is summarized below.

Definition 5 (Retraining DC) Consider the setting of Def. 4 with σ, Σ, β^* and sparsity target s . The sample $\hat{\beta}^{RT}$ from the retraining distribution $\mathcal{D}_{\sigma, \Sigma, \beta^*}^{RT, s}$ is constructed as follows. Sample $\hat{\beta} \sim \mathcal{D}_{\sigma, \Sigma, \beta^*}$ and compute the set of the top- s indices $\mathcal{I} = \mathcal{I}(\mathbb{T}_s(\hat{\beta}))$. Given $\mathcal{I}$, obtain the effective covariance $\Sigma_{\mathcal{I}} \in \mathbb{R}^{p \times p}$, population minima $\beta_{\mathcal{I}}^* \in \mathbb{R}^p$, and the noise level $\sigma_{\mathcal{I}} > 0$ as described above. Draw $\hat{\beta}^{RT} \sim \mathcal{D}_{\sigma_{\mathcal{I}}, \Sigma_{\mathcal{I}}, \beta_{\mathcal{I}}^*}$.

Observe that, the support $\mathcal{I}$ depends on the samples $\mathcal{S}$ via $\hat{\beta}$. Thus, our retraining DC is actually derived for the scenario when the retraining phase uses a fresh set of n samples to break the dependence between $\mathcal{I}, \mathcal{S}$ (which obtains $\hat{\beta}_{\text{fresh}}^{RT}$). Despite this, we empirically observe that the retraining DC predicts the regular retraining (reusing $\mathcal{S}$) performance remarkably well and perfectly predicts $\hat{\beta}_{\text{fresh}}^{RT}$ as discussed in Figs. 2 and 4. Finally, we defer the formalization of the retraining analysis to a future work. This includes proving that $\hat{\beta}_{\text{fresh}}^{RT}$ obeys Def. 5 asymptotically as well as directly studying $\hat{\beta}^{RT}$ by capturing the impact of the $\mathcal{I}, \mathcal{S}$ dependency.

6 Conclusions and Future Directions

This paper sheds light on under-explored phenomena in pruning practices for neural network model compression. On a theoretical level, we prove an accurate distributional

characterization (DC) for the solution of overparameterized least-squares for linear models with correlated Gaussian features. Our DC allows to precisely characterize the pruning performance of popular pruning methods, such as magnitude pruning. Importantly, our DC combined with a linear Gaussian equivalence, leads to precise analytic formulas for the pruning performance of nonlinear random feature models. On the experimental side, we provide a thorough study of overparameterization and pruning with experiments on linear models, random features and neural nets with growing complexity. Our experiments reveal striking phenomena such as a novel double descent behavior for model pruning and the power of overparameterization. They also shed light on common practices such as retraining after pruning.

Going forward, there are several exciting directions to pursue. First, it would be insightful to study whether same phenomena occur for other loss functions in particular for cross-entropy. Second, this work focuses on unregularized regression tasks and it is important to identify optimal regularization schemes for pruning purposes. For instance, should we use classical ℓ_1/ℓ_2 regularization or can we refine them by injecting problem priors such as covariance information? Finally, going beyond pruning, using DC, one can investigate other compression techniques that process the output of the initial overparameterized learning problem, such as model quantization and distillation.

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Potential Ethical Impacts

While deep learning is transformative in wide swath of applications, it comes at a cost: State-of-the-art deep learning models tend to be very large and consume significant energy during inference. The race for larger and better models and growing list of applications exacerbates this carbon footprint problem. Thus there is an urgent need for better and more principled model compression methods to help build environmentally friendly ML models. This work responds to this need by establishing the fundamental algorithmic principles and guarantees behind the contemporary model compression algorithms and by shedding light on the design of lightweight energy- and compute-efficient neural networks. We do not see an ethical concern associated with this work.

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